

Shahariar Asfaq Ovi

Machine Learning, Deep Learning & Backend AI Engineer | 2× ICPC Regionalist

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SUMMARY

Computer Science graduate specializing in Machine Learning, Deep Learning, and AI Engineering. Experienced in developing neural network architectures from scratch using NumPy, fine-tuning quantized LLMs using PEFT/LoRA, and building AI-powered applications with PyTorch and Python. Strong algorithmic problem-solving background as a two-time ICPC Dhaka Regional Finalist, complemented by practical backend engineering, data processing, and containerization experience. Interested in building scalable, production-ready machine learning solutions.

EDUCATION

University of Liberal Arts Bangladesh (ULAB) — **Dhaka, Bangladesh**

B.Sc. in Computer Science & Engineering (Completed) — *Jan 2022 — Sep 2026*

- **Academic Standing**: CGPA **3.10 / 4.00** (Graduated). Specializing in Artificial Intelligence, Backend AI Systems & Algorithms.

AWARDS & COMPETITIONS

Dhaka Regional Finalist | ICPC Dhaka Regional Contest — 2023, 2024

2nd Runners-up | Take-off Programming Contest Hosted By ULAB Computer Programming Club — 2023

Certificate of Participation | IUT 11th National ICT Fest — 2024

TECHNICAL SKILLS

Languages:	C++, Python, SQL, JavaScript, Java, PHP, Bash
Backend & APIs:	FastAPI, Flask, RESTful APIs, Asynchronous Python (asyncio), Uvicorn, PostgreSQL, API Integration
AI & LLM Systems:	Model Context Protocol (MCP), Agent Skills, Subagents, Claude API, Tool Calling, Agentic AI (ReAct), RAG, PEFT/LoRA
Data, Vision & Annotation:	Bounding Box Annotation, PDF Layout Analysis, Dataset Curation & RLHF, MediaPipe, WebAssembly, PyTorch, NumPy
Databases & DevOps Tools:	MySQL, SQLite, PostgreSQL, Git, GitHub Actions, Docker, Postman, Linux/Bash, Jupyter Notebook, LaTeX

PROFESSIONAL EXPERIENCE

FlyRank AI — **Remote**

Backend AI Engineer — Intern (Completed) — *Jun 2026 — Sep 2026*

- Architected an enterprise usage metering and quota enforcement engine using **FastAPI** and **PostgreSQL**, implementing exact integer micro-dollar pricing, guaranteed exactly-once idempotency deduplication, and automated Stripe webhook synchronization.
- Developed an automated LLM support ticket triage microservice with strict **Pydantic v2** schema validation, single-repair retry loops, quarantine error logging, and dynamic kill-switch controls.

GenMorphics AI Solution — **Dhaka, Bangladesh**

AI Data Trainer — *Jan 2025 — Nov 2025*

- Annotated and structured **1,000+ complex PDF documents** using high-precision bounding-box spatial labeling across tables, headers, and multi-column layouts to train multimodal OCR and Document Intelligence models.
- Evaluated and refined **10,000+** LLM prompt-response pairs, applying strict alignment heuristics and human-preference criteria (RLHF/DPO) to boost completion accuracy by **12%**.

ULAB Computer Programming Club — **Dhaka, Bangladesh**

Organizing Secretary — *Jan 2024 — Jan 2025*

- Directed club technical operations, managing intra-university competitive programming contests and mentoring 150+ student members in data structures & algorithms.

TECHNICAL PROJECTS

Project ROAR — Multi-Agent Intelligent Tutoring System (ULAB Capstone) — **GitHub Repo**

Python, PyTorch, PEFT/LoRA, MCP 2.x, FastAPI, ChromaDB, Docker — *Jan 2026 — Sep 2026*

- Architected a 5-agent Hierarchical Multi-Agent System (HMAS) orchestrated via **Model Context Protocol (MCP 2.x)** JSON-RPC bus, eliminating cross-agent context drift across Onboarding, Lesson, Quiz, Evaluator, and Scout agents.
- Built **ChromaDB RAG pipeline** indexing 1,899 vetted QA chunks (reducing hallucination 4.1×) with a 36-node Bloom taxonomy curriculum DAG.
- Fine-tuned quantized (**4-bit NF4**) **DeepSeek-R1-Distill-Qwen-7B** via **PEFT/LoRA** for dual-stage evaluation (regex rubric + semantic judge), achieving **F_1 = 95.8%** and **Cohen's κ = 0.8963** grading alignment under **4.82 GB VRAM**.

Agentic AI Toolkit — Framework-Free ReAct Agent — **GitHub Repo**

Python, Ollama, AI Agents, Tool Calling — *Sep 2026*

- Built a framework-free ReAct (Reason + Act) loop in pure Python standard library for transparent step-by-step reasoning, integrating local Ollama LLMs with custom tool calling for file system I/O, mathematical computation, and text analysis.

Orion — Multi-Agent Corrective RAG (CRAG) — **GitHub Repo**

Python, Gemini API, Streamlit, Vector Search — *Dec 2025*

- Programmed a multi-agent research assistant implementing Corrective RAG loops with NumPy cosine similarity evaluation (0.75 threshold), reducing hallucinations by **35%** and maintaining sub-120ms retrieval latency.

AeroDraw — Edge AI Gesture Controller — **GitHub Repo**

React, Vite, MediaPipe, WebAssembly — *May 2026*

- Engineered a client-side computer vision app converting WebAssembly-based MediaPipe hand-tracking landmark vectors into air-drawing strokes and cursor triggers at **60 FPS** with zero cloud latency.

OmegaNet — Neural Framework from Scratch — **GitHub Repo**

Python, NumPy, Deep Learning — *Nov 2025*

- Developed a modular deep learning framework with vectorized backpropagation, categorical cross-entropy loss, and custom Adam optimizer in pure NumPy, achieving **98.4%** validation accuracy.

CERTIFICATIONS (CLICK TO VIEW VERIFIED CREDENTIAL)

- Backend AI Eng. – FlyRank
- MCP: Advanced Topics – Anthropic
- Claude Code in Action – Anthropic
- AI Fluency for Builders – Anthropic
- Building w/ Claude API – Anthropic
- Intro to Agent Skills – Anthropic
- Claude w/ AWS Bedrock – Anthropic
- Agentic AI – Hugging Face
- Intro to MCP – Anthropic
- Intro to Subagents – Anthropic
- Claude Code 101 – Anthropic
- Open Science 101 – NASA TOPS